

08-Evaluation-Architecture.md

Healthcare Clinical Intelligence Platform

Enterprise Evaluation Architecture

Version: 1.0

Status: Approved Design

Mission

Build a continuous evaluation system that:

- Detects hallucinations
 - Measures retrieval quality
 - Measures agent quality
 - Prevents regressions
 - Improves trust
 - Supports healthcare governance
 - Enables safe production deployment
-

Evaluation Philosophy

Bad:

```
User reports issue
↓
Fix manually
```

Good:

```
Every Query
↓
Evaluate
↓
Measure
↓
```

Track
↓
Improve

Evaluation Layers

Layer 1

Ingestion Evaluation

Layer 2

Retrieval Evaluation

Layer 3

Agent Evaluation

Layer 4

Response Evaluation

Layer 5

Human Evaluation

Layer 6

Production Monitoring

Evaluation Pipeline

```
graph TD; Dataset --> Run_Evaluation[Run Evaluation]; Run_Evaluation --> Generate_Metrics[Generate Metrics]; Generate_Metrics --> Compare_Baseline[Compare Baseline]; Compare_Baseline --> Regression_Check[Regression Check]; Regression_Check --> Approval;
```

Evaluation Frameworks

Primary:

- RAGAS
- DeepEval

Secondary:

- Custom Healthcare Evaluators

Ingestion Evaluation

Measure:

- Parse Success Rate
- OCR Accuracy
- Metadata Completeness
- Chunk Quality
- Ontology Mapping Accuracy
- Graph Extraction Accuracy

Success Targets

Parse Success > 99%

Metadata Completeness > 95%

Chunk Quality > 90%

OCR Accuracy > 95%

Retrieval Evaluation

Mission:

Evaluate evidence quality.

Metrics

Context Precision

Context Recall

Context Relevancy

Source Diversity

Retrieval Latency

Retrieval Cost

Context Precision

Question:

How much retrieved context is useful?

Target:

>90%

Context Recall

Question:

Did we retrieve all relevant evidence?

Target:

>85%

Source Diversity

Target:

Minimum 3 sources

Example:

Hospital SOP

PubMed

FDA

Retrieval Confidence Accuracy

Compare:

Predicted Confidence

vs

Human Judgment

Target:

>90%

GraphRAG Evaluation

Measure:

Graph Coverage

Relationship Accuracy

Entity Accuracy

Path Accuracy

Agent Evaluation

Evaluate each agent separately.

Planner Agent

Metrics:

Intent Accuracy

Source Selection Accuracy

Tool Selection Accuracy

Cost Efficiency

Retriever Agent

Metrics:

Evidence Coverage

Retrieval Success

Retry Rate

Latency

Validator Agent

Metrics:

Hallucination Detection

Evidence Validation

Conflict Detection

Safety Agent

Metrics:

Risk Detection

Emergency Detection

Escalation Accuracy

False Positive Rate

False Negative Rate

Response Agent

Metrics:

Answer Quality

Citation Accuracy

Readability

Completeness

Hallucination Evaluation

Mission:

Detect unsupported claims.

Metrics

Faithfulness

Groundedness

Citation Coverage

Evidence Alignment

Target

Hallucination Rate < 5%

Citation Evaluation

Measure:

Citation Presence

Citation Accuracy

Citation Relevance

Citation Freshness

Target

Citation Coverage > 95%

Healthcare Evaluation

Special Healthcare Metrics

Clinical Accuracy

Does answer align with:

- SOPs
 - Guidelines
 - Medical Evidence
-

Safety Accuracy

Does answer avoid harmful advice?

Evidence Strength

Classify:

High Evidence

Moderate Evidence

Low Evidence

Golden Dataset

Maintain:

Approved Benchmark Questions

Categories

Clinical

Research

Drug Interactions

Policies

Insurance

Patient Education

Emergency Cases

Size

V1:

500 Questions

Future:

5000+ Questions

Regression Testing

Every release:

Run Golden Dataset

↓

Compare Previous Version

↓

Approve / Reject

Human Review System

Sample:

5% of production queries

Review:

Answer Quality

Retrieval Quality

Safety

Citations

User Feedback Evaluation

Collect:

Helpful

Not Helpful

Hallucination

Wrong Citation

Incomplete

Outdated

Feedback Metrics

Helpfulness Rate

Feedback Rate

Negative Feedback Rate

Production Evaluation

Evaluate:

Every Query

Metrics

Latency

Cost

Confidence

Answer Quality

Retrieval Quality

Agent Performance

Drift Detection

Detect:

Knowledge Drift

Prompt Drift

Agent Drift

Retrieval Drift

Cost Evaluation

Track:

Cost Per Query

Cost Per Agent

Cost Per Retrieval

Cost Per Source

Target

Cost decreases over time

while quality increases.

Observability Integration

Every evaluation emits events.

Examples:

```
retrieval_evaluated  
answer_evaluated  
hallucination_detected  
safety_flag_triggered  
confidence_calculated
```

Evaluation Dashboard

Sections:

Ingestion

Retrieval

Agents

Safety

Hallucinations

Costs

Feedback

Production Health

Release Gates

Deployment blocked if:

Context Precision < 85%

Hallucination > 5%

Citation Coverage < 95%

Safety Accuracy < 95%

Continuous Improvement Loop

Feedback

↓

Evaluation

↓

Analysis

↓

Improvement

↓

Regression Testing

↓

Deployment

Success Targets

Context Precision > 90%

Context Recall > 85%

Hallucination Rate < 5%

Citation Coverage > 95%

Safety Accuracy > 95%

User Satisfaction > 4.5/5

Final Principle

If it cannot be measured, it cannot be trusted.

Evaluation is not a feature.

Evaluation is the safety system of the entire platform.